THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 8: **Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A sequence has $t_1 = 9$ and $t_{n+1} = t_n + 4$. Then t_6 equals: 1

(a) 28 **(b)** 29 **(c)** 30 **(d)** 31

2. The 12th term of the AP $40, 36, 32, 28, \ldots$ is: 1

(a) -5 **(b)** -4 **(c)** -3 **(d)** -2

3. The sum of the first 12 terms of the AP $7, 10, 13, 16, \ldots$ is: 1

(a) 270 **(b)** 294 **(c)** 292 **(d)** 282

4. The 7th term of the GP $3, 6, 12, \ldots$ is: 1

(a) 189 **(b)** 192 **(c)** 195 **(d)** 193

5. The number of terms in the AP $5, 12, 19, 26, \ldots$ ending at 180 is: 1

(a) 27 **(b)** 28 **(c)** 25 **(d)** 26

6. A sequence in which each term is a fixed multiple of the previous term is called: 1

(a) a triangular sequence **(b)** an alternating sequence
(c) a geometric progression **(d)** an arithmetic progression

7. If $t_n = 1n^2 + 7$, then t_9 is: 1

(a) 87 **(b)** 89 **(c)** 88 **(d)** 90

8. Which term of the sequence $t_n = 6n + 5$ is 89? 1

(a) 13th term **(b)** 14th term **(c)** 15th term **(d)** 16th term

9. A sequence has $t_1 = 15$ and $t_{n+1} = t_n + -2$. Then t_6 equals: 1

(a) 7 **(b)** 6 **(c)** 5 **(d)** 4

10. The 20th term of the AP $-6, -1, 4, 9, \ldots$ is: 1

(a) 90 **(b)** 89 **(c)** 88 **(d)** 91

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Show that $30, 26, 22, 18, \ldots$ is an AP and find its 9th term. 2

12. Write the first five terms of the sequence with $t_1 = 2$ and $t_{n+1} = 2t_n + 3$. 2

13. A sequence is defined by $t_n = 4n + 5$. Find t_6 and decide whether 65 is a term. 2

14. Show that $1, 9, 17, 25, \ldots$ is an AP and find its 15th term. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find which term of the AP $4, 11, 18, 25, \ldots$ equals 214. 3

16. Find the sum of the first 18 terms of the AP $9, 17, 25, 33, \ldots$. 3

17. Find the common ratio of the GP $6, 18, 54, \ldots$ and its 6th term. 3

OR

Find which term of the AP $150, 142, 134, 126, \ldots$ equals 6. 3

18. Find the sum of the first 11 terms of the AP $40, 34, 28, 22, \ldots$. 3

OR

Find the common ratio of the GP $1, 3, 9, \ldots$ and its 8th term. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A taxi charges a fixed ₹100 and ₹15 per km. 5

(i) Show that the fares for 1, 2, 3 km form an AP.

(ii) Find the fare for 30 km.

OR

Rohan saves ₹100 in the first month and ₹30 more each month than in the previous month. 5

(i) How much does he save in month 16?

(ii) How much does he save in 16 months in all?

20. A taxi charges a fixed ₹60 and ₹12 per km. 5

(i) Show that the fares for 1, 2, 3 km form an AP.

(ii) Find the fare for 25 km.

OR

Rohan saves ₹250 in the first month and ₹10 more each month than in the previous month. 5

(i) How much does he save in month 20?

(ii) How much does he save in 20 months in all?

END OF QUESTION PAPER — Total: 40 Marks

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